

Martin Trucchi

Computer science PhD student

École Polytechnique
✉ martin.trucchi@inria.fr

About me

I am a 1st year PhD computer science student at the LIX lab of the École Polytechnique, in France. I work under the supervision of Lutz STRASSBURGER, Yannick FORSTER and Dominik KIRST. I enjoy working on logic, computability theory, type theory, constructive mathematics and formalisation, with a particular interest about synthetic and reverse mathematics.

Studies

- 2022-2026 **Diploma of the ENS of Lyon**
Ongoing
- 2024-2025 **Master 2, LMFI**, Université Paris Cité
Logique Mathématique et Fondements de l'Informatique
(*MSC - Mathematical Logic and Foundations of Computer Science*)
 - Mathematics : proof theory, model theory, linear logic, type theory, category theory
 - Computer science : lambda calculus, computability theory, functional programming, proof assistants
- 2023-2024 **Master 1, ENS of Lyon**
Mention Informatique, parcours Informatique : concepts et applications
MSC-1st year : Computer science, concepts and applications
 - Complexity theory, optimisation, compilation, quantum computing, proof theory
- 2022-2023 **Bachelor's degree, ENS of Lyon**
Mention Informatique, parcours type Informatique Fondamentale
Computer science
 - Foundations of CS, programming theory, algorithms and complexity, logic
- 2020-2022 **CPGE**, Lycée Thiers, Marseille
MPSI 3 , MP **
- 2020 **A-level science, Mention Très Bien**, Lycée Chevreul Blancarde, Marseille

Experiences

Internships

- 2026 **Synthetic constructive algebras**, Department of computer science, Università di Verona
Five months research internship, supervised by Peter SCHUSTER
- 2025-2026 **Synthetic and constructive algorithmic randomness**, Inria Paris, Université Paris Cité
Three months research internship, supervised by Yannick FORSTER and Dominik KIRST.
- 2025 **Formalization of game semantics in Rocq**, LIS, Aix-Marseille Université
Four months research internship, supervised by Étienne MIQUEY and Pierre CLAIRAMBAULT.

- 2024 **Synthetic topology in Agda, a definitional quest**, *School of computer science*, University of Birmingham
Three months research internship, supervised by Vincent RAHLI.
- 2023 **Aléatoire algorithmique - *Algorithmic randomness***, *I2M*, Aix-Marseille Université
Six weeks research internship, supervised by Pierre GUILLON.

Schools

- 2026 **PS26 - Proof Society**, *Aussois, France*
Participation (student) at the PS26 school, the 8th International School and Workshop on Proof Theory
- 2025 **EPIT25 - École de Printemps d'Informatique Théorique**
Spring School of Theoretical Computer Science, *Aussois, France*
Participation (student) at the EPIT25 school, about (Co)inductive & circular reasoning applied to programming, formal proofs and software verification.
- 2025 **PC25 - Proofs and computation**, *Herrshing, Germany*
Participation (student) at the PC25 school, about Predicative foundations, constructive Mathematics and type theory, computation in higher types and extraction of programs from proofs.

Teaching

- 2023-2024 Oral examination and labs of computer science in prépa class MP2I, La Martinière - Monplaisir High school in Lyon.

Languages

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|----------|-------------------|----------------------------------|
| Mastered | Français, Anglais | LaTeX, Python, Ocaml, Rocq, Agda |
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